

DPT Exercises

Exercise sheet 7

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Exercise 1 (sum of independent Poissons is Poisson). Let $X \sim \text{Poi}(\lambda_1)$ and $Y \sim \text{Poi}(\lambda_2)$ be independent RVs. Show that $X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$.

[Hint: Compute the characteristic function of a Poisson RV and use properties of characteristic functions.]

Solution: The characteristic function of a Poisson random variable $X \sim \text{Poi}(\lambda)$ is $\phi_X(t) = \exp(\lambda(e^{it} - 1))$. To see this we compute

$$\begin{aligned}\phi_X(t) &= \mathbb{E}[e^{itX}] = \sum_{k=0}^{\infty} e^{itk} P(X = k) = \sum_{k=0}^{\infty} e^{itk} \frac{e^{-\lambda} \lambda^k}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^{it})^k}{k!} = e^{-\lambda} \exp(\lambda e^{it}) = \exp(\lambda e^{it} - \lambda) = \exp(\lambda(e^{it} - 1)).\end{aligned}$$

Since X_1 and X_2 are independent, we have for $S = X_1 + X_2$ that

$$\begin{aligned}\phi_S(t) &= \phi_{X_1}(t) \phi_{X_2}(t) \\ &= \exp(\lambda_1(e^{it} - 1)) \exp(\lambda_2(e^{it} - 1)) \\ &= \exp((\lambda_1 + \lambda_2)(e^{it} - 1)).\end{aligned}$$

This is the characteristic function of a Poisson distribution with parameter $\lambda_1 + \lambda_2$. By the uniqueness property of characteristic functions, we conclude that $S \sim \text{Poi}(\lambda_1 + \lambda_2)$.

Exercise 2 (server overload). A web server receives requests from two types of users: Type A and Type B. Let N_A be the number of requests from Type A users in an hour, with $\mathbb{E}[N_A] = 50$ and $\text{var}[N_A] = 50$. Similarly, N_B is the number of requests from Type B users in an hour, with $\mathbb{E}[N_B] = 100$ and $\text{var}[N_B] = 100$. Assume N_A and N_B are independent. The server has a capacity to handle $C = 200$ requests per hour. If the total number of requests $N_T = N_A + N_B$ exceeds C , the server is overloaded and breaks down.

- (i) Calculate the expected total number of requests per hour, $\mathbb{E}[N_T]$, and the variance of the total number of requests per hour, $\text{var}[N_T]$.
- (ii) Using Chebyshev's inequality, provide an upper bound on the probability that the server is overloaded, i.e., on $P(N_T > C)$.
- (iii) Assuming N_T is non-negative (quite plausibly), use Markov's inequality and $\mathbb{E}[N_T]$ to provide another upper bound on $P(N_T > C)$. Compare it to the bound in (ii) and explain the difference.
- (iv) What would be an adequate distribution to model N_A , N_B and why?
- (v) One can show that for $X \sim \text{Poi}(\lambda_1)$, $Y \sim \text{Poi}(\lambda_2)$ and $X \perp\!\!\!\perp Y$, we have that $X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$. Using this result, calculate the probability of server overload $P(N_T > C)$ under this model. (You can use a computer for this. For example, `scipy.stats` has functions for the cdfs of most common distributions.) Compare this value with the bounds obtained in (ii) and (iii). Discuss the differences and the utility of the bounds versus an exact calculation for a specific distribution.

Solution:

- (i) We use the linearity of the expectation to compute $\mathbb{E}[N_T] = \mathbb{E}[N_A + N_B] = \mathbb{E}[N_A] + \mathbb{E}[N_B] = 50 + 100 = 150$. Since $N_A \perp\!\!\!\perp N_B$, the variance is also additive and we find $\text{var}[N_T] = \text{var}[N_A + N_B] = \text{var}[N_A] + \text{var}[N_B] = 50 + 100 = 150$.

- (ii) We want to bound $P(N_T > 200)$. Since N_T is number of requests, it's discrete, so $P(N_T > 200) = P(N_T \geq 201)$. This is $P(N_T - \mathbb{E}[N_T] \geq 201 - 150) = P(N_T - 150 \geq 51)$. Since $P(N_T - 150 \geq 51) \leq P(|N_T - 150| \geq 51)$, we can use Chebyshev's inequality with $\epsilon = 51$. $P(|N_T - 150| \geq 51) \leq \frac{\text{var}[N_T]}{\epsilon^2} = \frac{150}{51^2} = \frac{150}{2601} \approx 0.05767$. So we conclude that the probability of a server overload is at most about 5.77%.
- (iii) As the number of requests, N_T is non-negative. The Markov inequality is simply $P(N_T > 200) \leq \frac{\mathbb{E}[N_T]}{200} = 0.75$. The Chebyshev bound ($\approx 5.77\%$) is *much tighter* than the Markov bound, which says that the probability of an overload is at most 75%. This is expected because Chebyshev's inequality uses both the mean and the variance, providing more information about the distribution's spread, while Markov's inequality only uses the mean (and non-negativity).
- (iv) A Poisson distribution is adequate for modeling N_A and N_B . This is because counts of events occurring independently in a fixed time interval are often well-described by a Poisson distribution, and for a Poisson random variable $X \sim \text{Poi}(\lambda)$, we have $\mathbb{E}[X] = \lambda$ and $\text{var}[X] = \lambda$ (perfectly fitting the exercise statement). So, $N_A \sim \text{Poi}(50)$ and $N_B \sim \text{Poi}(100)$.
- (v) If $N_A \sim \text{Poi}(50)$ and $N_B \sim \text{Poi}(100)$ are independent, then $N_T = N_A + N_B \sim \text{Poi}(50 + 100) = \text{Poi}(150)$. We want $P(N_T > 200) = 1 - P(N_T \leq 200) = 1 - \sum_{k=0}^{200} \frac{e^{-150} 150^k}{k!}$. Using, e.g., Python with `1 - scipy.stats.poisson.cdf(200, 150)` we get $P(N_T > 200) \approx 0.000042 = 0.0042\%$. The probability of overload under the Poisson model is orders of magnitudes smaller than the Chebyshev and Markov bounds, which are essentially distribution-free. They are super general but often not very tight. While the Poisson model may make us very confident about our server uptime, it likely underestimates the risk of requests not coming in at a fixed rate independently from users A and B. There is again a clear tradeoff between getting exact results under concrete model assumptions versus (potentially loose) bounds when not committing to such assumptions.

Exercise 3. Let $(\Omega, \mathcal{A}, P)$ be a probability space, and (X, Y) a $\mathbb{R}^2$ -valued RV that is uniformly distributed on the disc

$$\overline{D^2} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

The density of this uniform distribution is constant on the disc and has value π , since the area of the unit disc is π . That is,

$$p_{X,Y}(x, y) = \frac{1}{\pi} \chi_{\overline{D^2}}(x, y).$$

- (i) Compute the marginal densities $p_X(x)$ and $p_Y(y)$.
- (ii) Compute the means and variances $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\text{var}[X]$, $\text{var}[Y]$.
[Hint: For $\mathbb{E}[X^2]$, consider the substitution $x = \sin(u)$ and look up trigonometric identities for $\sin(2u)$ as well as for expressing $\sin(x)^2$ in terms of a cos. Use the symmetry between X and Y .]
- (iii) Check whether $X \perp\!\!\!\perp Y$ and whether X and Y are uncorrelated.
[Hint: Use polar coordinates to perform the integral for the covariance. Remember that you have to transform the integration boundaries and add the Jacobian determinant in the integral.]

Solution:

- (i) As $\overline{D^2} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ we have $p_X(x) = 0$ for $x \notin [-1, 1]$. For $x \in [-1, 1]$ we compute

$$p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x, y) dy = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{2}{\pi} \sqrt{1-x^2}.$$

Hence $p_X(x) = \frac{2}{\pi} \sqrt{1-x^2} \chi_{[-1,1]}(x)$. By the symmetry of X and Y , we have equivalently $p_Y(y) = \frac{2}{\pi} \sqrt{1-y^2} \chi_{[-1,1]}(y)$.

(ii) For the mean $\mathbb{E}[X]$, we calculate

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x p_X(x) dx = \frac{2}{\pi} \int_{-1}^1 \underbrace{x}_{\text{odd}} \underbrace{\sqrt{1-x^2}}_{\text{even}} dx = 0.$$

Note that x is an odd function (when flipped around the x-axis, a minus sign is required to make it equal again) whereas $\sqrt{1-x^2}$ is an even function (symmetric around the y-axis). Hence, the product of them is an odd function (“even · odd = odd”). Any integral with symmetric boundaries (from $-a$ to a) over an odd function vanishes (under some regularity conditions).

Similarly, with the substitution $x = \sin(u)$ (such that $\frac{dx}{du} = \cos(u)$), we compute

$$\mathbb{E}[X^2] = \frac{2}{\pi} \int_{-1}^1 x^2 \sqrt{1-x^2} dx = \frac{4}{\pi} \int_{-1}^1 x^2 \sqrt{1-x^2} dx = \frac{1}{\pi} \int_0^{\pi/2} \underbrace{(2 \sin(u) \cos(u))^2}_{=\sin(2u)} du$$

and using $\sin^2(x) = \frac{1}{2} - \frac{1}{2} \cos(2x)$ gives

$$\mathbb{E}[X^2] = \frac{1}{\pi} \int_0^{\pi/2} \left(\frac{1}{2} - \frac{1}{2} \cos(4u) \right) du = \frac{1}{\pi} \left(\frac{u}{2} - \frac{1}{8} \sin(4u) \right) \Big|_0^{\pi/2} = \frac{1}{\pi} \cdot \frac{\pi}{4} = \frac{1}{4}.$$

Thus, we obtain

$$\text{var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{1}{4}.$$

(iii) To check for correlation, we use polar coordinates

$$\Psi(r, \theta) = \begin{pmatrix} r \cos(\theta) \\ r \sin(\theta) \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \Psi^{-1}(\overline{D^2}) = [0, 1] \times [0, 2\pi).$$

The Jacobi-determinant of Ψ is $\det(J_\Psi(r, \theta)) = r$ and we obtain

$$\begin{aligned} \text{cov}[X, Y] &= \int_{\mathbb{R}^2} xy p_{X,Y}(x, y) d(x, y) = \int_{\overline{D^2}} \frac{xy}{\pi} d(x, y) = \int_0^{2\pi} \int_0^1 r \cos(\theta) r \sin(\theta) \frac{1}{\pi} r dr d\theta \\ &= \frac{1}{2\pi} \int_0^1 r^3 \underbrace{\int_0^{2\pi} \sin(2\theta) d\theta}_{=0} dr = 0. \end{aligned}$$

Therefore, X and Y are uncorrelated. However, X and Y are not independent: for any $(x, y) \in (-1, 1)^2 \setminus \overline{D^2}$ we have $p_{X,Y}(x, y) = 0$, but

$$p_X(x)p_Y(y) = \left(\frac{2}{\pi} \sqrt{1-x^2} \chi_{[-1,1]}(x) \right) \left(\frac{2}{\pi} \sqrt{1-y^2} \chi_{[-1,1]}(y) \right) \neq 0 \quad \text{for all } (x, y) \in (-1, 1)^2.$$

Exercise 4 (coding: Monte Carlo (MC) integration). The laws of large numbers (we will see those formally in the lecture) imply that we can *estimate* the mean $\mathbb{E}[Y]$ of an RVRV Y by “averaging n iid samples,” that is

$$\mu := \mathbb{E}[Y] \approx \frac{1}{n} \sum_{i=1}^n Y_i.$$

This can be used for numerical “quadrature,” that is, for estimating integrals $I := \int_U f(x) dx$ over some bounded domain $U \subsetneq \mathbb{R}^d$. By generating/sampling samples $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(U)$ (that is, all X_i are iid uniform), the estimate using the actual instantiations $x_1, \dots, x_n$ reads

$$\hat{I}_n := \frac{1}{n} \sum_{i=1}^n f(x_i) \approx \frac{1}{\lambda^d(U)} \int_U f(x) dx.$$

Use this concept to numerically approximate the integral of $f(x) := 10 - x_1^2 - x_2^2$ over the unit disc

$$D^2 = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 < 1\}$$

by implementing the following steps:

- (i) Draw n samples uniformly independently inside D^2 . (Make sure they are in D^2 and not $(-1, 1)^2$.)
- (ii) Estimate the measure of D^2 with these samples in a suitable way, that is, can you use these samples to estimate the area of $D^2 = \pi$.
- (iii) Then use the samples to approximate the integral $\int_{D^2} f(x) dx$. Compare this to the true analytical value of the integral.

Solution: For a solution, see the Jupyter notebook. Compare your obtained integral result with the integral

$$\iint_{D^2} (10 - x_1^2 - x_2^2) dx_1 dx_2 = \int_0^1 \int_0^{2\pi} (10 - r^2) r d\varphi dr = 2\pi \cdot \left(5r^2 - \frac{r^3}{3}\right) \Big|_0^1 = 2\pi \frac{14}{3} \approx 29.34.$$

Optional computation rules for Gaussians

We are not expecting you to actually go through all of these, especially the later ones become quite tedious. The idea is just that you've seen the solutions and how these computations are being performed. Note that we will provide "direct computations" in the solutions, i.e., computing directly with pdfs and cdfs. For many of those computations, cfs are extremely useful and make computations much more concise.

Exercise 5. The pdf of a multi-variate Gaussian with mean $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ is given by

$$p(x) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)\right),$$

where $|\Sigma|$ denotes the determinant of Σ . We then write $x \sim \mathcal{N}(\mu, \Sigma)$. Any valid covariance matrix has to be symmetric ($\Sigma^T = \Sigma$) and positive definite, meaning that $x^T \Sigma x > 0$ for all $x \in \mathbb{R}^d \setminus \{0\}$. We also denote positive definiteness by $\Sigma \succ 0$. Let $X \in \mathbb{R}^{d_X}, Y \in \mathbb{R}^{d_Y}$ be random vectors that are jointly distributed according to a multi-variate Gaussian with mean $\mu = (\mu_X, \mu_Y) \in \mathbb{R}^d$ for $d := d_X + d_Y$ and covariance matrix

$$\Sigma = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_{YY} \end{pmatrix} \in \mathbb{R}^{d \times d},$$

where $\Sigma_{XX} \in \mathbb{R}^{d_X \times d_X}$, $\Sigma_{XY} \in \mathbb{R}^{d_X \times d_Y}$, and $\Sigma_{YY} \in \mathbb{R}^{d_Y \times d_Y}$. Calling $Z := (X, Y)$ we thus have $z \sim \mathcal{N}(\mu, \Sigma)$. Prove the following statements:

- (i) The random variables X and Y are independent if and only if $\Sigma_{XY} = 0$.
- (ii) The random variable $AZ + b$ for $A \in \mathbb{R}^{d \times d}$ and $b \in \mathbb{R}^d$ follows the multi-variate normal $\mathcal{N}(A\mu + b, A\Sigma A^T)$. (You can assume that $A\Sigma A^T \succ 0$.)
- (iii) The marginal distribution of X is given by $\mathcal{N}(\mu_X, \Sigma_{XX})$ (and the marginal distribution of Y is given by $\mathcal{N}(\mu_Y, \Sigma_{YY})$).
- (iv) The conditional distribution of X given $Y = y$ is given by $\mathcal{N}(\mu_{X|y}, \Sigma_{X|y})$ where

$$\mu_{X|y} = \mu_X + \Sigma_{XY} \Sigma_{YY}^{-1} (y - \mu_Y), \quad \Sigma_{X|y} = \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{XY}^T.$$

- (v) If we assume $d_X = d_Y$ and $\Sigma_{XY} = 0$, i.e., the random vectors X and Y are over the same space and are uncorrelated and thus independent. Then the random variable $aX + bY$ is distributed according to $\mathcal{N}(a\mu_X + b\mu_Y, a^2\Sigma_{XX} + b^2\Sigma_{YY})$ for $a, b \in \mathbb{R}$.
- (vi) The previous statement can be extended: Even in the original case ($d_X \neq d_Y$ and $\Sigma_{XY} \neq 0$), we have that for all matrices $A \in \mathbb{R}^{k \times d_X}$, $B \in \mathbb{R}^{k \times d_Y}$ and vectors $c \in \mathbb{R}^k$, the random vector $AX + BY + c$ is distributed according to multi-variate normal in k dimensions given by

$$\mathcal{N}(A\mu_X + B\mu_Y + c, A\Sigma_{XX}A^T + B\Sigma_{YY}B^T + A\Sigma_{XY}B^T + B\Sigma_{YX}A^T).$$

(Again, the resulting normal may be “degenerate”, meaning that the covariance matrix may not be positive definite depending on A and B .)

Solution: We first start by showing that a normally distributed, multidimensional RV $X : (\Omega, \mathcal{A}, P) \rightarrow \mathbb{R}^{d_X}$, such that $P_X \sim \mathcal{N}(\mu_X, \Sigma_{XX})$ satisfies $\mathbb{E}[X] = \mu_X$. This follows as

$$\begin{aligned} \mathbb{E}[X] &= \frac{1}{\sqrt{(2\pi)^{d_X} |\Sigma_{XX}|}} \int_{\mathbb{R}^{d_X}} x \exp\left(-\frac{1}{2}(x - \mu_X)^\top \Sigma_{XX}^{-1}(x - \mu_X)\right) dx \\ &= \mu_X \underbrace{\frac{1}{\sqrt{(2\pi)^{d_X} |\Sigma_{XX}|}} \int_{\mathbb{R}^{d_X}} \exp\left(-\frac{1}{2}u^\top \Sigma_{XX}^{-1}u\right) du}_{=1} + \underbrace{\frac{1}{\sqrt{(2\pi)^{d_X} |\Sigma_{XX}|}} \int_{\mathbb{R}^{d_X}} u \exp\left(-\frac{1}{2}u^\top \Sigma_{XX}^{-1}u\right) du}_{=0} \\ &= \mu_X \end{aligned}$$

where in the first step, we just inserted the above probability density function and in the second step, we substituted $u = x - \mu_X$. To see the last step, we first have to note that Σ is symmetric (and positive definite), hence by the spectral theorem diagonalizable with $\Sigma = U\Lambda U^\top$, $U \in O(n)$ such that $\Sigma^{-1} = U\Lambda U^\top = U\Lambda^{-1/2}\Lambda^{-1/2}U^\top$. Thus for $v = uU\Lambda^{-1/2}$, we obtain for the first term, since $\det \Lambda^{1/2} = \det \Sigma_{XX}^{1/2}$

$$= \frac{1}{\sqrt{(2\pi)^{d_X}}} \int_{\mathbb{R}^{d_X}} \exp\left(-\frac{1}{2}v^\top v\right) dv = \frac{1}{\sqrt{(2\pi)^{d_X}}} \iiint \prod_{i=1}^{d_X} \exp\left(-\frac{1}{2}v_i^2\right) dv_1 \dots dv_{d_X} = 1$$

as $\int_{\mathbb{R}} \exp\left(-\frac{1}{2}v_i^2\right) dv_i = \sqrt{2\pi}$. The second term is calculated similar leveraging that

$$\int_{-\infty}^{\infty} v_i \exp\left(-\frac{1}{2}v_i^2\right) dv_i = \left[\exp\left(-\frac{1}{2}v_i^2\right)\right]_{-\infty}^{\infty} = 0.$$

In addition, we show that for $Z = (X, Y)^\top$ satisfies

$$\text{cov}[Z] = \begin{pmatrix} \text{cov}[X] & \text{cov}[X, Y] \\ \text{cov}[Y, X] & \text{cov}[Y] \end{pmatrix} = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{XY}^\top & \Sigma_{YY} \end{pmatrix}.$$

Similar as above

$$\begin{aligned} \text{cov}[Z] &= \mathbb{E}\left[(Z - \mu_Z)(Z - \mu_Z)^\top\right] \\ &= \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \int_{\mathbb{R}^d} (z - \mu_Z)(z - \mu_Z)^\top \exp\left(-\frac{1}{2}(z - \mu_Z)^\top \Sigma^{-1}(z - \mu_Z)\right) dz. \end{aligned}$$

Again, as Σ is symmetric, by the spectral theorem, it is diagonalizable as $\Sigma = U\Lambda U^\top$ with $U \in O(n)$ orthogonal matrix Λ diagonal matrix and notice that $\Sigma^{-1} = U\Lambda U^\top = U\Lambda^{-1/2}\Lambda^{-1/2}U^\top = B^\top B$. Hence with substitution $u = B(z - \mu_Z)$, (hence $u^\top = (z - \mu_Z)^\top B^\top$, $u^\top B^{-\top} = (z - \mu_Z)^\top$) the above formula becomes

$$= \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \int_{\mathbb{R}^d} B^{-1}uu^\top B^{-\top} \exp\left(-\frac{1}{2}u^\top u\right) |B| du = B^{-1} \underbrace{\text{cov}[U, U]}_{I_{d \times d}} B^{-\top} = \Sigma.$$

Here, in the last step we used the fact that by integration by parts

$$\int_{-\infty}^{\infty} x^2 \exp(-\frac{1}{2}x^2) dx = x \exp(-\frac{1}{2}x^2) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -\exp(-\frac{1}{2}x^2) dx = \sqrt{2\pi}$$

(i) " $\Rightarrow$ ": Assume $X \perp\!\!\!\perp Y$. Then it holds $\text{cov}[X, Y] = \Sigma_{XY} = 0$.

" $\Leftarrow$ ": Given $\Sigma_{XY} = 0$. Then we can write

$$\begin{aligned} p_Z(z) &= \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp \left(-\frac{1}{2} ((x - \mu_X)^\top \Sigma^{XX} (x - \mu_X) + (y - \mu_Y)^\top \Sigma^{YY} (y - \mu_Y)) \right) \\ &= \frac{1}{\sqrt{(2\pi)^{d_X} |\Sigma_{XX}|}} \exp \left(-\frac{1}{2} (x - \mu_X)^\top \Sigma^{XX} (x - \mu_X) \right) \\ &\quad \cdot \frac{1}{\sqrt{(2\pi)^{d_Y} |\Sigma_{YY}|}} \exp \left(-\frac{1}{2} (y - \mu_Y)^\top \Sigma^{YY} (y - \mu_Y) \right) \\ &= p_X(x) p_Y(y) \end{aligned}$$

where we have used $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - CB)$.

(ii) We have given the linear map (assumed bijective) $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $f(z) = Az + b$ (bijective here means A full rank). Then we have

$$P(U \in A) = \int_A p_U(u) du$$

as well as, using the change-of-variable formula for the push-forward measure

$$P(U \in A) = P(Z \in f^{-1}(A)) = \int_{f^{-1}(A)} p_Z(z) dz = \int_A p_Z(f^{-1}(u)) |\det Df^{-1}(u)| du$$

thus giving

$$p_U(u) = p_Z(f^{-1}(u)) |\det Df^{-1}(u)|.$$

Therefore

$$\begin{aligned} p_U(u) &= \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp \left(-\frac{1}{2} (A^{-1}(u - b) - A^{-1}(A\mu))^\top \Sigma^{-1} (A^{-1}(u - b) - A^{-1}(A\mu)) \right) |A^{-1}| \\ &= \frac{1}{\sqrt{(2\pi)^d |A\Sigma A^\top|}} \exp \left(-\frac{1}{2} (u - (b + A\mu))^\top (A\Sigma A^\top) (u - (b + A\mu)) \right). \end{aligned}$$

In case A is not injective, we use the pseudo-inverse $A^\dagger$ the inverse of $A|_{\ker^\perp}$ (or moment-generating functions).

(iii) For $Z = \begin{pmatrix} X \\ Y \end{pmatrix}$, $\mu = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}$ and $\Sigma = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{XY}^\top & \Sigma_{YY} \end{pmatrix}$. We then just apply the previous result for $A = [Id_{d_X}, 0]^\top$ and $b = 0 \in \mathbb{R}^{d_X}$. Then we obtain

$$X = AZ + b \sim \mathcal{N}(\mu_X, \Sigma_{XX})$$

(iv) Assume we are given $Z = \begin{pmatrix} X \\ Y \end{pmatrix}$, $\mu = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}$ and $\Sigma = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{XY}^\top & \Sigma_{YY} \end{pmatrix}$ with $Z \sim \mathcal{N}(\mu, \Sigma)$, $Y \sim \mathcal{N}(\mu_Y, \Sigma_{YY})$. Then

$$\begin{aligned} p(x | y) &= \frac{p(x, y)}{p(y)} = \frac{((2\pi)^d |\Sigma|)^{-1/2}}{((2\pi)^{d_Y})^{-1/2}} \exp \left(-\frac{1}{2} (z - \mu)^\top \Sigma^{-1} (z - \mu) + \frac{1}{2} (y - \mu_Y)^\top \Sigma_{YY}^{-1} (y - \mu_Y) \right) \\ &= c \exp \left(-\frac{1}{2} \left((x - \mu_X)^\top M_{xx} (x - \mu_X) + 2(x - \mu_X)^\top M_{xy} (y - \mu_Y) + (y - \mu_Y)^\top (\Sigma_{yy}^{-1} - M_{yy}) (y - \mu_Y) \right) \right) \end{aligned}$$

where $c = \frac{((2\pi)^d |\Sigma|)^{-1/2}}{((2\pi)^{d_Y})^{-1/2}} = \frac{1}{\sqrt{(2\pi)^{d_X} |\Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx}|}}$ used for abbreviation. Using the matrix identity $\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} (A - BD^{-1}C)^{-1} & (A - BD^{-1}C)^{-1}BD^{-1} \\ -D^{-1}C(A - BD^{-1}C)^{-1} & D^{-1} + D^{-1}C(A - BD^{-1}C)^{-1}BD^{-1} \end{pmatrix}$, we hence obtain

$$\begin{aligned} &= c \exp \left(-\frac{1}{2} \left((x - \mu_x)^\top (\Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx}) (x - \mu_x) - 2(x - \mu_x)^\top (\Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx}) \Sigma_{xy} \Sigma_{yy}^{-1} (y - \mu_y) \right. \right. \\ &\quad \left. \left. + (y - \mu_y)^\top \Sigma_{yy}^{-1} \Sigma_{yx} (\Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx})^{-1} \Sigma_{xy} \Sigma_{yy}^{-1} (y - \mu_y) \right) \right) \\ &= c \exp \left(-\frac{1}{2} \left(x - (\mu_x + \Sigma_{xy} \Sigma_{yy}^{-1} (y - \mu_y)) \right)^\top (\Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx})^{-1} \left(x - (\mu_x + \Sigma_{xy} \Sigma_{yy}^{-1} (y - \mu_y)) \right) \right) \end{aligned}$$

which is what we wanted to show.

(v) We first assume for simplicity that $d_x = d_y = 1$. Then it holds

$$P(Z \leq z) = P(X + Y \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y) dy dx = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y) dy dx$$

where the last step followed from the independence. Then:

$$\begin{aligned} f_Z(z) &= \frac{\partial}{\partial z} F_Z(z) = \int_{-\infty}^{\infty} f_Y(z - x) f_X(x) dx = \frac{1}{2\pi \sigma_x \sigma_y} \int_{-\infty}^{\infty} \exp \left(-\left(\frac{(z - x - \mu_y)^2}{2\sigma_y^2} + \frac{(x - \mu_x)^2}{2\sigma_x^2} \right) \right) dx \\ &= c \int_{-\infty}^{\infty} \exp \left(-\frac{x^2(\sigma_x^2 + \sigma_y^2) - 2x(\sigma_x^2(z - \mu_y) + \sigma_y^2 \mu_x) + \sigma_x^2(z^2 + \mu_y^2 - 2z\mu_y) + \sigma_y^2 \mu_x^2}{2\sigma_x^2 \sigma_y^2} \right) dx \end{aligned}$$

with $c = \frac{1}{(\sqrt{2\pi})^2 \sigma_z \sigma_x \sigma_y / 2}$ with $\sigma_z = \sqrt{\sigma_x^2 + \sigma_y^2}$. Therefore

$$\begin{aligned} &= c \int_{-\infty}^{\infty} \exp \left(\frac{x^2 - 2x \left(\frac{\sigma_x^2(z - \mu_x) + \sigma_y^2 \mu_x}{\sigma_z^2} \right) + \left(\frac{\sigma_x^2(z^2 + \mu_y^2 - 2z\mu_y) + \sigma_y^2 \mu_x^2}{\sigma_z^2} \right)}{2 \left(\frac{\sigma_x \sigma_y}{\sigma_z} \right)^2} \right) dx \\ &= \frac{1}{\sqrt{2\pi} \sigma_z} \exp \left(\underbrace{-\frac{\sigma_z^2(\sigma_x^2(z - \mu_x)^2 + \sigma_y^2 \mu_x^2) - (\sigma_x^2(z - \mu_y) + \sigma_y^2 \mu_x)^2}{2\sigma_z^2(\sigma_x \sigma_y)^2}}_{= -\frac{(z + (\mu_x + \mu_y))^2}{2\sigma_z^2}} \right) \cdot \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \frac{\sigma_x \sigma_y}{\sigma_z}} \exp \left(-\frac{(x - A)^2}{2 \left(\frac{\sigma_x \sigma_y}{\sigma_z} \right)^2} \right) dx}_{=1} \end{aligned}$$

Thus we obtain

$$\Rightarrow Z \sim \mathcal{N}(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$$

With the choice $X' = aX \sim \mathcal{N}(a\mu_X, a^2\sigma_X^2)$, $Y' = bY \sim \mathcal{N}(b\mu_Y, b^2\sigma_Y^2)$ we obtain $Z = X' + Y' \sim \mathcal{N}(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$

(vi) By assumption $\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma)$ (X, Y jointly Gaussian) and assume for simplicity $d_X = d_Y = 1$.

Then we obtain from (iii) that

$$\begin{pmatrix} X \\ W \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{\rho \sigma_Y}{\sigma_X} & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N} \left(\tilde{\mu}, \begin{pmatrix} 1 & -\frac{\rho \sigma_Y}{\sigma_X} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{\rho \sigma_Y}{\sigma_X} & 1 \end{pmatrix} \right) = \mathcal{N} \left(\tilde{\mu}, \begin{pmatrix} \sigma_X^2 & 0 \\ 0 & \sigma_Y^2(1 - \rho^2) \end{pmatrix} \right)$$

with $\tilde{\mu} = \begin{pmatrix} \mu_X \\ \mu_Y - \frac{\rho\sigma_Y}{\sigma_X}\mu_X \end{pmatrix}$. Hence we obtain from (v) that

$$\begin{aligned}
aX + bY &= \left(aX + b\frac{\rho\sigma_Y}{\sigma_X} \right) X + bW \\
&\sim \mathcal{N} \left(\left(a + b\frac{\rho\sigma_Y}{\sigma_X} \right) \mu_X + b \left(\mu_Y - \frac{\rho\sigma_Y}{\sigma_X} \mu_X \right), \left(a + b\frac{\rho\sigma_Y}{\sigma_X} \right)^2 \sigma_X^2 + b^2(1 - \rho^2)\sigma_Y^2 \right) \\
&= \mathcal{N} (a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2, a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y)
\end{aligned}$$